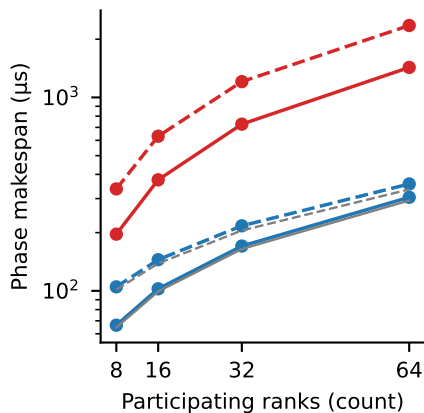


Collective width tail: void study with findings

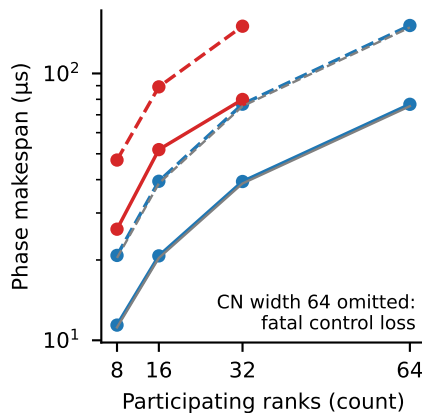
64-rank reference; physical two-tier Clos at 400 / 200 Gbit/s

— NN: mic-nn ideal — NN phase floor (a, b) - - - 200 Gbit/s
— CN: mic-cn physical — 400 Gbit/s

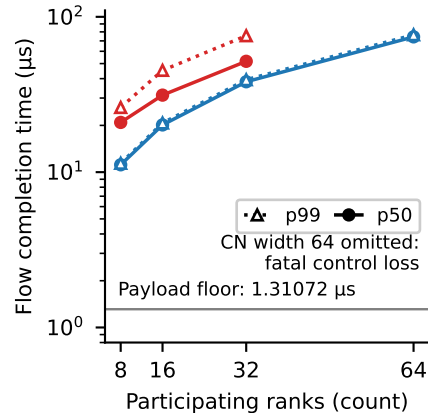
(a) Ring all-reduce
Phase makespan



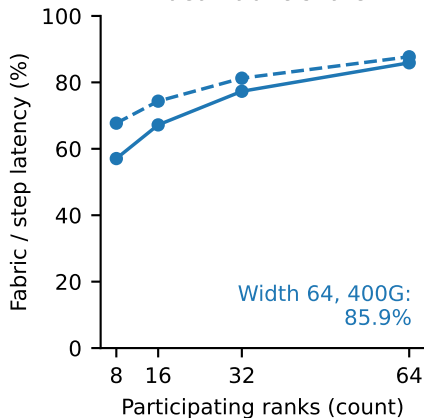
(b) Remote all-to-all
Phase makespan



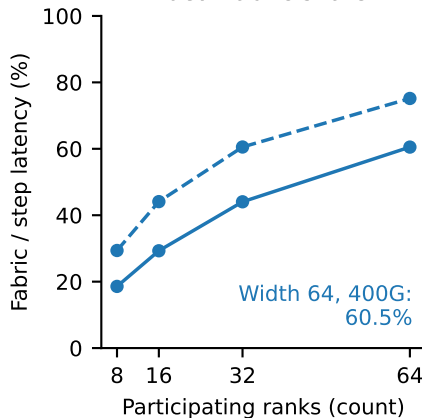
(c) All-to-all flow tail
400 Gbit/s only



(d) Two-ring step
Ideal fabric share



(e) Single-engine expert step
Ideal fabric share



(f) Phase makespan ratio
Physical / ideal

